

Using LLMs to Score News and Opinion Media Based on Their Past Predictions

Jane Bambauer¹, Andi Bettinger², Abhiram Durga Gorle³, Venkat Lokesh Vejendla⁴, Jim Harper⁵

Introduction

We introduce a theory, methodology, and early prototype of an LLM tool that can provide a novel form of media source quality using old predictions. We show that an LLM tool can identify old predictions, gauge the verifiability of the prediction, identify a certainty level and “due date” for the predictions, and then grade them. If this tool performs well enough and at scale, it can provide a credible reputation score for journalists, public intellectuals, and politicians that augments existing news quality metrics.

Background and Motivation

There is increasing evidence that “misinformation” problems are at least as much a demand-side phenomenon as they are supply-side ones. When disinformation, misinformation, and bullshit goes viral, it is often because people who already are inclined to believe the claim want to see it. [1,2,3] Fact-checking was a once-promising attempt to curb problems of political misperception by providing a signal of quality for specific claims, articles, or sources of information.

These efforts have had some success, but they are fallible, slow, and distrusted. [4, 5,6] One of the greatest challenges to fact-checking is that the signal is often produced only after a particular fact has become *contested*. By attempting to intervene in a live controversy, some proponents on one side of the debate will update not their belief in the fact, but their trust in the fact-checking institution. Moreover, some fact-checking organizations such as NewsGuard puts undue weight on the procedures, expert trust, and past reputation of a news source which skeptics will understandably believe has biased its scores in favor of legacy media.

Nevertheless, the creation of valid and broadly trusted reputation signals is a worthy goal. Speaker reputation is very important because most knowledge is built from the testimony of others rather than from direct experiences and observations. [7,8] A broadly-trusted and well-functioning reputation score will help steer not only the intrinsically truth-seeking members of the audience but even the typical news consumer by making trust in low-quality media sources more socially embarrassing and psychologically taxing.

Our Work

We are building LLM-based tools to judge the quality of the media in new ways that we hope can be more trusted and trustworthy than fact-checking. These include: (a) logic-checking; (b) cherry-pick-checking; (c) “fluff”-detection (that is, measuring the proportion of material that provides neither evidence nor argument); and (d) prediction-scoring.

This article addresses our tool designed to grade old predictions.

¹ Eminent Scholar, University of Florida College of Journalism & Communications and Levin College of Law

² University of Florida

³ University of Florida (M.S. in Computer Science, expected 2026)

⁴ University of Florida (M.S. in Computer Science, expected 2026)

⁵ University of Florida and American Enterprise Institute

How it works:

Our tool uses fine-tuned LLMs to:

1. Identify *gradable* prediction
2. Identify conditions that must be met, if any, as a premise to the prediction
3. Define verifying future events
4. Define falsifying future events
5. Select a deadline based on verifying and falsifying events
6. Attribute the claim to speakers/sources
7. Grade (binary) the prediction when the deadline has passed
8. Score (weighted by confidence) the prediction
9. Combine prediction scores into appropriate source quality scores

A limited public prototype of our tool is available for readers to play with at <https://newsgrader.streamlit.app/>. Ultimately, our goal is to scan news stories, opinion pieces, and podcast transcripts at scale and run them through the grader so that information sources and frequent commentators can be judged based on a large enough compilation of old predictions.

The potential payoff:

Prediction-grading sidesteps some of the problems inherent to fact-checking because there is usually little debate about whether a prediction has or has not been met. This is the reasons that prediction markets are able to clear without constant controversy. No existing system, to our knowledge, automatically discovers, extracts, and scores the accuracy of predictions over time to create performance profiles for authors and publications. Prior to the prevalence of LLM technologies, it was not possible to automate the tracking and scoring of predictions. Although prediction scoring alone is not sufficient to get a holistic measure of a speaker's or source's epistemic quality, it can provide a signal to supplement other formal and informal tools that audiences use to assess quality and assign trust. If it works well enough, prediction scores can influence speaker and audience behavior and can provide a tool for researchers that complements NewsGuard scores.

However, the utility of our tool will ultimately depend on the soundness of using predictions as a form of quality and the performance of LLMs in accurately identifying and assessing predictions.

Predictive Performance as Source Quality

A speaker's or publication's track record of predictive accuracy is one objective and transparent measure of news quality that avoids the risks of political bias and false balance (that is, the risk that comes from insisting on representation of all views when some are inconsistent with the facts.) By focusing exclusively on verifiable predictions, our method sidesteps the controversies of fact-checking and bias labeling, offering a data-driven and objective alternative for holding public voices accountable. To be clear, we do not mean to suggest that this is the *only* quality measure, or even the best one, but it can provide one credible signal for news quality that doesn't require a fact-checker to decide a contentious issue.

Right now, punditry and commentary suffer from a "cheap talk" environment. A pundit can make a bold claim and safely ignore it without reputational consequence if it turns out to be wrong. Prediction-tracking, if widely adopted as a signal, can change the environment by closing the memory hole. Authors and experts can be chosen based (partly) on their accuracy in past forecasts rather than on pure

credentialism or experience. Moreover, a prediction scoring system could have a disciplining impact on prospective behavior if a speaker will know that their claims and speculations will be logged and later checked against reality. If the grading mechanism is designed well, speakers will be rewarded not only for making more accurate predictions, but for stating their claims with more precision and care.

Humans are naturally competitive and love games with clear rules and scores. [9] Prediction tracking turns truth-seeking into a sport. Tracking predictions transforms truth-seeking from a private virtue into a public, measurable, and gamified skill. It helps foster a meritocracy of expertise where status is awarded not for loyalty, loudness, or confidence, but for a demonstrable track record of being correct.

Ideally, prediction-scoring can also reward the best pundits with social capital by making the act of careful, honest reasoning a high-status or cool pursuit. People often rely on peripheral cues, such as source prestige, rather than engaging in central processing and critical evaluation of arguments when evaluating information. [10, 11] Tracking prediction accuracy provides a quantifiable metric that serves as a peripheral cue for credibility. This aligns with findings that expertise identified before message delivery increases persuasion. [12, 13, 14] The prediction scoring we use in this project also rewards calibration—that is, how well a speaker’s confidence matches their accuracy. [15, 16, 17] Individuals who accurately assess their knowledge limits are better forecasters. Making calibration visible can reward intellectual humility and truth-tracking, and can deter blind faith in certain narratives and worldviews.

Research on reputation systems find that providing information about aggregated performance metrics significantly increased cooperation. [18] Reputation systems (like star ratings on eBay or Yelp) are crucial for establishing trust in environments where people interact without prior knowledge of each other, but not all information is processed equally. Disaggregated data and individual ratings are cognitively demanding to process for the bounded rationality of humans. [19, 20] Aggregating data into a simple, salient score can dramatically alter behavior by making reputation information easier to understand and more actionable. Aggregated performance metrics (usually the average) are also more fair than individual disaggregated cases since individual cases may give the wrong impression about a speaker’s overall quality.

Our work builds on and learns from several existing research areas:

1. Quantified Accuracy

Other scholars have created systems for automating fact-checking at scale. Systems like ClaimBuster aim to automate the identification of "check-worthy" claims. [22] Our work is similarly interested in scaling up the capacity for news quality assessments. Their NLP methods for claim detection are useful and have some similarities to our efforts to identify predictions. However, our system is more likely to run into the ground truth problem when facts are currently contested. Predictions may be contested on partisan grounds when they are made, but not when they are resolved. Although it’s possible that some predictions will be difficult to resolve because of partisan divisions over what wound up happening after the prediction was made, this problem is likely to afflict just a small subset of gradable predictions.

2. Prediction Markets

A substantial body of research has demonstrated that aggregate forecasts, often in the form of prediction markets, can aggregate useful distributed information and provide accurate forecasts. [23, 24, 25, 26] Platforms like Polymarket and Kalshi allow users to trade shares based on the outcome of real-world events, creating a market price that reflects the crowd's collective probability estimate for that event. Empirical studies have consistently shown that these markets frequently outperform individual expert

predictions in domains such as politics and economics. [23, 27] The concept of prediction markets, specifically political future markets, dates back to as early as 1503. [28] Thus, we have long understood that markets in predictions might create the right incentives to think objectively and reveal latent information.

Prediction markets work when there is a high participation rate among people with relevant, dispersed information. The share value synthesizes the private information of the participants into a single, constantly updating probability. This collective intelligence has been shown to frequently outperform the best individual experts as well as the collective assessment of people who are not financially rewarded for success. Social desirability bias can undermine the wisdom of crowds, but as economist Eric Zitzewitz put it, “there’s no virtue-signaling in an anonymous market when you’re betting.” You are financially rewarded for being correct, not for being popular, ideologically consistent, or politically correct. [29]

Prediction markets do not generally indicate which individuals are consistently accurate. Some “superforecasters” trumpet their own accomplishments but generally markets like Polymarket are anonymous. Anonymity may help the accuracy of the market by inducing more people to participate, even if they want to maintain their privacy. But the main reason prediction markets are expected to work well is because the people participating have “skin in the game”—cheap talk is no longer cheap if you lose money from it. [30] News media, experts, and influencers could have skin in the game, too, if their reputations were gauged based on how accurate their predictions turned out to be, but nobody keeps track of their performance, at least not at any scale.

Political psychologist Philip Tetlock has led groundbreaking research exploring whether experts’ predictions are significantly better than lay persons’ predictions [15] and also demonstrating that ordinary people could be organized to make remarkably accurate forecasts about geopolitical events. [31] The “Good Judgment Project” (GJP) operates as a reputation market, where a forecaster’s track record is their most valuable asset. The project found that while the “wisdom of crowds” performed better than a typical person making predictions alone, some individuals (“superforecasters”) are consistently and significantly more accurate than others. Forecasting is a measurable skill. The GJP proved that accuracy can be tracked and attributed to individuals, and a reputation for good judgment can be built empirically.

Tetlock’s methods couldn’t be done at scale without the aid of LLMs. Our system runs the “Good Judgment Project” in reverse. Our project does for journalists and pundits what Tetlock did for forecasters, making their track record legible. If this can be done, reputation scores will create a feedback loop where being proven wrong has a consequence for one’s credibility and being proven right builds trust. Readers could actively seek out analysts with high “accuracy scores,” moving beyond brand loyalty or ideological alignment to a more evidence-based form of trust.

3. The Forecasters Dilemma

Our work also builds on scholarship that solves the “forecaster’s dilemma.” This is a phenomenon in which forecasts on extreme and unlikely outcomes (like financial crises, natural disasters, etc) can lead to distortions when evaluating the accuracy of predictions. [32] Skillful forecasts may appear worse than they are, and poor or “extremist” forecasts (e.g., always predicting disaster) may appear better than they are if the scoring system is not set up to account for them. We will discuss how prediction scores should be calculated below. But it is worth noting that intuitive assessments of prediction performance is likely to be worse than almost any formal system because extreme and unlikely events are the ones that attract the most public attention. If forecasters are judged based on predicting these events, doomsayers will be valorized and good forecasters will be penalized.

LLM Competence Generally

Even if prediction-scoring is a viable way to change behavior through reputation, are LLMs competent to accurately grade performance? We suspect so. LLMs like GPT-4 outperform humans in predicting the next word in a sequence, *especially* in contexts where the semantic cues are ambiguous or weak, a scenario analogous to identifying non-explicit predictions. [21] Zhang and Strelnikov found that the AI's advantage was most pronounced in 'low-predictability' contexts. [21] In today's information environment, predictions are often implied, and also hedged with language like "could," "might," or "is poised to." They are often embedded in complex sentences. Yet, as past research and our own experience suggests, LLMs appear to pick up nonobvious contextual cues as well as humans.

Zhang and Strelnikov did find that AI can generate "plausible but unconventional" completions. This means we should expect that LLM models will have Type I (false positive) and Type II (false negative) error in the process of classifying predictions: some predictions will be missed, and some non-predictions will be identified. As we explain below, we found that these errors were primarily the result of inherent brittleness in the nature of grammar and human communications, where listeners are highly sensitive to how information is phrased. In other words, we would expect that human reviewers will disagree about which statements should count as gradable predictions. We explain below how we handled this using multiple LLM agents to create a conservative pool of predictions, where a prediction is only counted and graded if it multiple models considered the statement to be a prediction.

Our Methods and Results

After an initial manual phase of exploration and several iterations with three LLM models (OpenAI, Gemini, and Grok), we determined the necessary components of a prediction scoring system.

Each prediction entry must:

- Identify a gradable prediction
- Identify conditions, if any
- Define verifying event(s)
- Define falsifying event(s)
- Select deadline(s) based on verifying and falsifying events
- Attribute claims across speakers and outside sources
- Gauge the confidence in the prediction for each speaker
- Grade the prediction (binary)
- Score the prediction (weighted by confidence)
- Assimilate the score into the speakers' prediction quality metric (which combines scores across multiple predictions)

We will explain the meaning and operationalization of each step as well as current results to our testing. We also created three datasets constructed from a stratified randomized selection of news and opinion article from which we created training and testing datasets. In the first dataset, every sentence was hand-coded three times by human coders. For the second and third datasets, the subset of sentences initially selected by our baseline model (which had previously had 100% recall—that is, no false negatives) were hand-coded by humans.

1. *Identifying Gradable Predictions* (statement about future, objectively falsifiable)

First, the tool scans the text to find language making a gradable prediction. A *prediction* is a statement indicating an anticipated future state. A prediction is *gradable* if it can be objectively tested for truth or falsity once the relevant time arrives. It should be specific, measurable, and timebound.⁶

Specific: Is it clear, in context, what would need to happen or not happen in order to verify or falsify the prediction? This will be necessary for the “verifying and falsifying events” stage below.

Measurable: Will evidence become available to test the verifying or falsifying conditions? This will be necessary for grading the prediction.

Timebound: In context, is there a reasonable deadline for evaluating the verifying or falsifying conditions? This will be necessary to find the deadline that triggers the grading.

Operationalizing this definition is, therefore, the major challenge of our work. Identifying gradable predictions requires the model to take into account the nuances of context and language. For example, the word “will” does not always signify a prediction. Examples of false positive (incorrectly identified predictions) include:

And she prayed, which she says she'll continue to do.
(example of “will” as a description of present intent)

Mamdani will take office on January 1, 2026.
(example of “will” as a description of present plans)

Republicans will continue to grapple with losses during Tuesday night’s elections.
(example of ungradable vagueness of verifying or falsifying event (“grapple”))

Western civilization is doomed.
(example of ungradable vagueness of deadline)

We found numerous examples of mirage predictions in our article databases. Nevertheless, even news articles routinely trade in gradable predictions, and political podcasts and opinion-makers use them even more frequently. Throughout this essay, we will use the following three examples of gradable predictions to illustrate our tool:

[A] *Rep. Adelita Grijalva cannot be seated*

Context:

AOC has definitely grown on me but she needs to work on her legislative prowess in order to become the complete package. Congressional Chronicle: Week of October 6, 2025. The House is still adjourned and Speaker Johnson will not reconvene it. That means *Rep. Adelita Grijalva cannot be seated* — and it is impossible to avoid a prolonged shutdown.

⁶ Ungradable predictions can be excluded altogether from the accuracy score. But they can also reflect a speaker’s tendency to engage in bluster and vagueness instead of precise content. A different tool under development, the “fluff detector,” will keep track of speakers’ frequent use of ungradable predictions.

[B] *It is impossible to avoid a prolonged shutdown*

Context:

AOC has definitely grown on me but she needs to work on her legislative prowess in order to become the complete package. Congressional Chronicle: Week of October 6, 2025. The House is still adjourned and Speaker Johnson will not reconvene it. That means Rep. Adelita Grijalva cannot be seated — and *it is impossible to avoid a prolonged shutdown*.

[C] *It was necessary if the school is going to avoid tough sanctions from the federal Department of Justice Civil Rights Division*

Context:

At a meeting on Friday, part open and part closed to the public, the board voted to eliminate a variety of diversity, equity and inclusion programs that had become fronts for discrimination in admissions and hiring at the school. This is a rebuke to Gregory Washington, the George Mason president, a vociferous advocate of "diversity" preferences who had long opposed the anti-DEI move. But *it was necessary if the school is going to avoid tough sanctions from the federal Department of Justice Civil Rights Division*. The DOJ is investigating the school's policies for possible violations.

In Appendix A, we provide several categories of rhetorical or linguistic phrases that can trick an untrained LLM into misidentifying a prediction.

We were pleasantly surprised to find that even zero-shot models produced less error than we thought they would. The models we tested—Open AI's GPT-5, Gemini, and Grok—varied in their responses (as did the human testers), but each of them independently was quite good.⁷ We created a version 1 of the tool (the "Baseline") using GPT-5 and triangulating the grading across the other models without any finetuning or further training.

Since version 1, we have tested seven refined models that use small (3-shot or 7-shot) or large (BERT Model) amounts of training data, that use backtracking based on later-assessed variables (such as the verifying or falsifying events), or that used Chain of Thought (CoT) reasoning. Below is our table of performance metrics:

	Precision	Recall	F1
Baseline (0-shot)	0.288	1.000	0.447
3-shot	0.4402	0.9419	0.6
7-shot	0.7203	0.9884	0.8333
Backtracking	0.7917	0.6032	0.6847
Modified Backtracking	0.8182	0.7778	0.7975
Hybrid Backtracking with Rubric	0.7385	0.6154	0.6713
Chain of Thought (CoT) Hybrid	0.6458	0.5636	0.6019
BERT Model	0.8223	0.7447	0.7801

⁷ We also used Llama but found it to be inferior to the other three, for now.

The baseline model had total recall, but it was so chockfull of false positives that precision was poor. The modified backtracking model has the highest F1 score, but at present, our preferred model is the BERT model. We prioritize precision over recall for the sake of caution. A tool that misses some gradable predictions is still useful, but a tool that wrongly attributes predictions to speakers and sources who can legitimately challenge the categorization will produce noise and distrust. Indeed, of all the steps in tool-building, we consider isolating predictions to the exclusion of non-predictions to be the most crucial and challenging step. We are not yet sure whether 0.82 precision is sufficient. (It may be if downstream grading can further purge the false positives without human oversight.)

2. Identifying Conditions on the Predictions, if Any

Conditional predictions establish a logical dependency between a trigger event and an outcome. They're not standalone forecasts but contingent claims. This matters because in volatile domains like geopolitics or economics, conditional statements often represent the most sophisticated analysis. These statements do not predict the conditions, only the consequences if the predictions are met. Thus, conditional predictions require special tracking and will only be graded if the conditions occurred.

Using our three examples, the second prediction "...and it is impossible to avoid a prolonged shutdown" is, in context, conditional on the previous predictions coming true. Of all the possible reasons that a shutdown could be prolonged, the speaker is only claiming one chain of causation: namely that *if* the house stays adjourned, and *if* Adelita Grijalva is not seated, *then* the shutdown will be prolonged.

Ordinarily, conditional predictions are graded only if the conditions are met. This example, however, raises some interesting scoring challenges that we will discuss below.

3. Setting the Prediction's Implied Verifying or Falsifying Events

Polymarket and other prediction markets can work without complaints and disputes overwhelming the system by specifying in advance a deadline and a binary condition on which the prediction will be deemed correct. Predictions made out in the wild do not have the same disciplining structure, but they very often imply events and deadlines that can be inferred by LLMs.

For each prediction, our tool is forced to elaborate the specific events that would prove the prediction correct, and specific events that would prove the prediction false. Each set of events, whether verifying or falsifying, are disjunctive sets (meaning that they should be described by the model such that if any one of the events occurs, the claim is verified/falsified). Sometimes, the superset of verifying and falsifying events covers the universe of possibilities. In this case, the prediction is completely gradable. It will be proven true or false. Other times, the prediction is only partially gradable because although it can be verified or disproven, there are some possible events in the future that a reasonable observer would not feel confident categorizing as verifying or falsifying the claim. We preserve these partially gradeable predictions and grade them when a verifying or falsifying event occurs, and not otherwise.

Let's return to our examples to see how this works in practice. Here are the outputs for our three examples:

[A] *Rep. Adelita Grijalva cannot be seated*

Gradeability	Verifying Events	Falsifying Events	Gradeability Notes
Gradable	• House remains adjourned preventing swearing-in ceremony • Clerk of the House confirms no quorum or session for seating • Rep. Grijalva's seat officially remains vacant in records	• Speaker reconvenes House allowing swearing-in to proceed • Emergency session held specifically to seat new members • Legal mechanism found to seat her without full House session	Directly verifiable — congressional records confirm seating status. Binary outcome.

[B] *It is impossible to avoid a prolonged shutdown*

Gradeability	Verifying Events	Falsifying Events	Gradeability Notes
Partially Gradable	• Shutdown extends past 2+ weeks with no deal announced • Both parties publicly signal no imminent resolution • Federal agencies begin furloughing non-essential workers en masse	• Continuing resolution passed within days ending the shutdown • Emergency bipartisan deal reached before major disruptions materialize • Presidential emergency powers invoked to fund government temporarily	"Prolonged" is the ambiguity — needs an operational definition (e.g., >2 weeks). The shutdown itself is binary and verifiable.

[C] *It was necessary if the school is going to avoid tough sanctions from the federal Department of Justice Civil Rights Division*

Gradeability	Verifying Events	Falsifying Events	Gradeability Notes
Partially Gradable	• DOJ investigation concludes and explicitly cites DEI elimination as a mitigating factor • George Mason avoids sanctions while similar schools that kept DEI programs face penalties • DOJ issues formal letter acknowledging the board's compliance efforts	• DOJ sanctions George Mason despite DEI program elimination • DOJ closes investigation with no sanctions regardless of DEI status • Federal courts block DOJ enforcement, making compliance moot	"Necessary" is causal — hard to prove definitively. Gradeability depends on DOJ report language. Deadline tied to expected report release.

4. Setting Deadline(s) Based on Verifying and Falsifying Events

A prediction without an implicit deadline is ungradeable because it can be indefinitely deferred. A prediction must have a boundary after which we can declare it true or false. Thus, the elusiveness of a deadline is a major source of vagueness and a reason that some predictions are deemed ungradeable.

A strict requirement for an explicit due date would disqualify the vast majority of predictions made by pundits, politicians, and commentators, who heavily rely on vague timeframes like "soon," "after the election," or "in the coming years." To avoid losing a large volume of predictions, we instruct the LLMs to use the verifying and falsifying events and the broader context to set a conservative (as in uncontroversial) deadline.

Here is how the tool handled our three examples (with the verifying and falsifying events reproduced):

[A] *Rep. Adelita Grijalva cannot be seated*

Verifying Events	Falsifying Events	Deadline	Gradeability Notes
• House remains adjourned preventing swearing-in ceremony • Clerk of the House confirms no quorum or session for seating • Rep. Grijalva's seat officially remains vacant in records	• Speaker reconvenes House allowing swearing-in to proceed • Emergency session held specifically to seat new members • Legal mechanism found to seat her without full House session	~1–2 weeks	Directly verifiable — congressional records confirm seating status. Binary outcome.

[B] *It is impossible to avoid a prolonged shutdown*

Verifying Events	Falsifying Events	Deadline	Gradeability Notes
• Shutdown extends past 2+ weeks with no deal announced • Both parties publicly signal no imminent resolution • Federal agencies begin furloughing non-essential workers en masse	• Continuing resolution passed within days ending the shutdown • Emergency bipartisan deal reached before major disruptions materialize • Presidential emergency powers invoked to fund government temporarily	2–4 weeks	"Prolonged" is the ambiguity — needs an operational definition (e.g., >2 weeks). The shutdown itself is binary and verifiable.

[C] *It was necessary if the school is going to avoid tough sanctions from the federal Department of Justice Civil Rights Division*

Verifying Events	Falsifying Events	Deadline	Gradeability Notes
• DOJ investigation concludes and explicitly cites DEI elimination as a mitigating factor • George Mason avoids sanctions while similar schools that kept DEI programs face penalties • DOJ issues formal letter acknowledging the board's compliance efforts	• DOJ sanctions George Mason despite DEI program elimination • DOJ closes investigation with no sanctions regardless of DEI status • Federal courts block DOJ enforcement, making compliance moot	3–6 months	"Necessary" is causal — hard to prove definitively. Gradeability depends on DOJ report language. Deadline tied to expected report release.

The entries for example [C], which we include for full transparency, contains an error. One of the falsifying events suggests that the prediction would be false if the DOJ sanctions George Mason despite DEI program elimination. In fact, the prediction suggests that program removal is a necessary-but-not-necessarily-sufficient condition. Thus, the proper falsifying events are the second and third bullet point, but not the first. This will be relevant during grading since the first “falsifying” event is what in fact occurred.

5. *Attributing Predictions to Multiple Speakers*

Authors of a story often quote or paraphrase other people in the course of publishing a prediction, and this raises ambiguity about whether it is fair to judge the author according to predictions that come from an interviewee or some other external speaker.

Sometimes, the prediction is presented in a manner that makes clear that the source of the prediction and the author/publisher are independent voices. This is most clear when an author/publisher has quoted a prediction in order to explicitly disagree with it. Consider the following:

Lucy Miller says that Rosebud will win the Kentucky Derby but that’s preposterous. Rosebud was ill last month and won’t be fit enough to win.

In this case, the author’s criticism of the source’s prediction clarifies that the author is not making Miller’s prediction. In fact, the criticism creates a mirror-opposite prediction that is attributable to the author. As a result, this particular passage has two predictions: one in which Lucy Miller is the source (that Rosebud will win) and one in which the author is the source (that Rosebud will NOT win.)

Most of the time, thought, the external speaker’s prediction is not criticized, and the prediction should be attributed to the author as well as the external speaker. After all, when an author decides to quote another person, the author is implicitly signaling "this prediction is important enough for you to hear." The audience perceives the prediction as credible precisely because the publication has chosen to repeat it. So for every prediction, our tool attempts to create attribution data so that grades and scores can be calculated or weighted differently depending on whether the author is “platforming” and implicitly boosting the external speaker’s prediction or not.

6. Gauging Confidence

The confidence dimension captures two phenomena that are not worth separating: the level of conviction expressed by the speaker (e.g. “I think Rosebud will win the Kentucky Derby, but I’m not sure) and the level of probability that the speaker adopts, often implicitly (e.g. “I am highly confident that Rosebud has a 41% chance of winning the Kentucky Derby, and the next closest horse in my estimation has only a 12% chance of winning.”) The latter represents high confidence in a probability model while the former may have no particular probability model in mind. For example, contrast your prediction of whether the stock market will go up or down tomorrow with whether you think a coin will flip heads. For anybody who is not a day trader, the probability model for the stock market is vague and unknown, while we have complete confidence that there is a 50% chance of flipping heads.

While it would be useful to keep track of both of these meanings of confidence¹, we ultimately treat all predictions as probabilistic forecasts.

We plan to instruct the LLM model to either to assign a probability (p) attached to the prediction based on the content and full context of the speech, or to use a 5-point confidence scale often used for calculating Brier scores. [34] We have just begun to do the fine-tuning on this part of our system. If the prediction is attributed to multiple speakers (e.g. a quoted expert as well as the author), the confidence scores assigned to each speaker can vary.

7. Grading the Prediction

If the verifying and falsifying events are set up well, and if gradeability notes took into account the likely sources or means to resolve the prediction, the grading stage should be straightforward. Partially gradeable events may be proved true, false, or ungraded (in which case the prediction is removed.) If there are reasons to doubt the veracity or completeness of the sources that would verify the prediction, this, too, would be a reason to remove the prediction *ex post*.

Here is how our tool graded the three examples:

[A] *Rep. Adelita Grijalva cannot be seated*

Gradeability	Gradeability Notes	Verdict	Justification
Gradable	Directly verifiable — congressional records confirm seating status. Binary outcome.	TRUE	Grijalva was denied seating for 50 days after winning her Sept 23 special election. Johnson refused to swear her in, citing the shutdown. She was finally seated on Nov 12, 2025 only after the House reconvened. The prediction was accurate during the window it was made.

¹ When a speaker expresses a prediction with high certainty and is proven correct, it demonstrates both knowledge and confidence. This is more impressive and informative than a correct prediction made with low certainty. Conversely, when a speaker is highly certain but wrong, it indicates overconfidence and possibly poor judgment. This should be penalized more heavily than being wrong with low certainty.

[B] *It is impossible to avoid a prolonged shutdown*

Gradeability	Gradeability Notes	Verdict	Justification
Partially Gradable	"Prolonged" is the ambiguity — needs an operational definition (e.g., >2 weeks). The shutdown itself is binary and verifiable.	TRUE	The shutdown lasted a record 43 days (Oct 1–Nov 12, 2025), the longest full government shutdown in US history. It surpassed the prior 35-day record set in 2018–2019. By any reasonable definition of 'prolonged,' this prediction was clearly correct.

[C] *It was necessary if the school is going to avoid tough sanctions from the federal Department of Justice Civil Rights Division*

Gradeability	Gradeability Notes	Verdict	Justification
Partially Gradable	"Necessary" is causal — hard to prove definitively. Gradeability depends on DOJ report language. Deadline tied to expected report release.	MIXED	The DEI elimination did not prevent sanctions — in August 2025, the Dept of Education found GMU in violation of Title VI regardless, ordering a Resolution Agreement. The DOJ launched a second investigation in July 2025 specifically naming President Washington. The DEI elimination was deemed 'cosmetic' by investigators. The prediction's causal logic (elimination = avoiding sanctions) did not hold.

As noted above, example [C] should be graded as neither true nor false rather than “mixed” because neither of the two proper falsifying conditions were met. Of course, since GMU was sanctioned despite removing the DEI programs, the clearly verifying conditions were also not met.

We have not yet fine-tuned the grading portion of our model, nor have we subjected this part to external validation and performance metrics. Stay tuned.

8. *Scoring the Prediction and Creating Reputation Score(s)*

Scoring the predictions takes into account that speakers often make prediction not because they are wholly confident that it will come true, but because they think it is more likely to come true than their audience might think—more likely than the audience’s priors. If people generally assume that there is only a 5% chance that the Iranian regime will collapse within the next year, but the speaker thinks the chance is closer to 30%, the speaker is likely to say “the mullahs still might fall.”

Suppose, in context, the prediction-grading tool is able to infer that the speaker predicts regime change in the next year with 30% confidence (that is, that it is 30% likely to occur.) If the event doesn’t occur, was 30% a bad one? If regime change does occur, was 30% a good prediction?

With a very large collection of predictions, a person’s performance could be assessed based on its calibration. Perfect calibration would mean that if you collected all of the 30% predictions, about 30% of them would occur and 70% would not. And if all of the 40% predictions were grouped together, 40% of them would materialize, too. Same for 50%, 60%, etc.

The Brier score is another option that we tentatively prefer. The Brier score incorporates calibration but also rewards correctness (assigning higher probability to events that occur).

$$\text{Brier Score} = (\text{Outcome} - p)^2$$

Where p is the speaker's confidence and Outcome is 1 if the event happened, 0 if it did not. A lower score is better, as perfect and perfectly confident predictions would result in 0.

Finally, reputation scores should be calculated for content creators, experts, and news organizations by aggregating the information collected through the procedures described above. But aggregating the predictions is not a straightforward process. Augmented scoring rules such as the Penalized Brier Score (PBS) and Penalized Logarithmic Loss (PLL) can make corrections to ensure that classifications are always scored more favorably than incorrect ones. [35]

Using a single aggregation system across all predictions is reasonable first step, but it will have unintended consequences and produce perverse incentives. If each prediction is weighed the same regardless of content, the system will reward sources who traffic in the banal (e.g. "temperatures will be higher in the summer"). In future work, we will consider how to disambiguate easy or obvious predictions from more meaningful ones so that the latter can be given greater weight.

Another potential problem is that the same speaker may make essentially the same prediction several times, both in the same work and across multiple works. Initially, we thought we would need to train the AI tool to check for substantive similarity and only track and grade the prediction one time. However, on further reflection, we are inclined to treat each instance of the prediction as its own instance. This way, a fabulist or conspiracy theorist or demagogue who repeats the same wrong claims over and over again is penalized in proportion to the amount of energy they put into that particular claim. And conversely, a Cassandra-like speaker who has tried multiple times to get the audience to see the light gets rewarded in proportion to their efforts to break through. As long as the problem of banal predictions is minimized through appropriate weighting, and as long as the scored source has made a varied enough set of predictions overall (e.g. at least 20 substantively unique predictions), grading all instances of the same or similar prediction is not a problem in our view.

Conclusion

Large Language Model technologies may open promising new paths to assess news quality with scale, speed, and methods that were not previously available. This project provides one example of such a tool and demonstrates that LLM models may be able to be used effectively with minimal re-training and fine-tuning.

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